

Cigarette Smoking and Pancreatic Cancer: A Pooled Analysis From the Pancreatic Cancer Cohort Consortium

Smoking is an established risk factor for pancreatic cancer; however, detailed examination of the association of smoking intensity, smoking duration, and cumulative smoking dose with pancreatic cancer is limited. The authors analyzed pooled data from the international Pancreatic Cancer Cohort Consortium nested case-control study (1,481 cases, 1,539 controls). Odds ratios and 95% confidence intervals were calculated by using unconditional logistic regression. Smoking intensity effects were examined with an excess odds ratio model that was linear in pack-years and exponential in cigarettes smoked per day and its square. When compared with never smokers, current smokers had a significantly elevated risk (odds ratio (OR) = 1.77, 95% confidence interval (CI): 1.38, 2.26). Risk increased significantly with greater intensity (≥ 30 cigarettes/day: OR = 1.75, 95% CI: 1.27, 2.42), duration (≥ 50 years: OR = 2.13, 95% CI: 1.25, 3.62), and cumulative smoking dose (≥ 40 pack-years: OR = 1.78, 95% CI: 1.35, 2.34). Risk more than 15 years after smoking cessation was similar to that for never smokers. Estimates of excess odds ratio per pack-year declined with increasing intensity, suggesting greater risk for total exposure delivered at lower intensity for longer duration than for higher intensity for shorter duration. This finding and the decline in risk after smoking cessation suggest that smoking has a late-stage effect on pancreatic carcinogenesis.